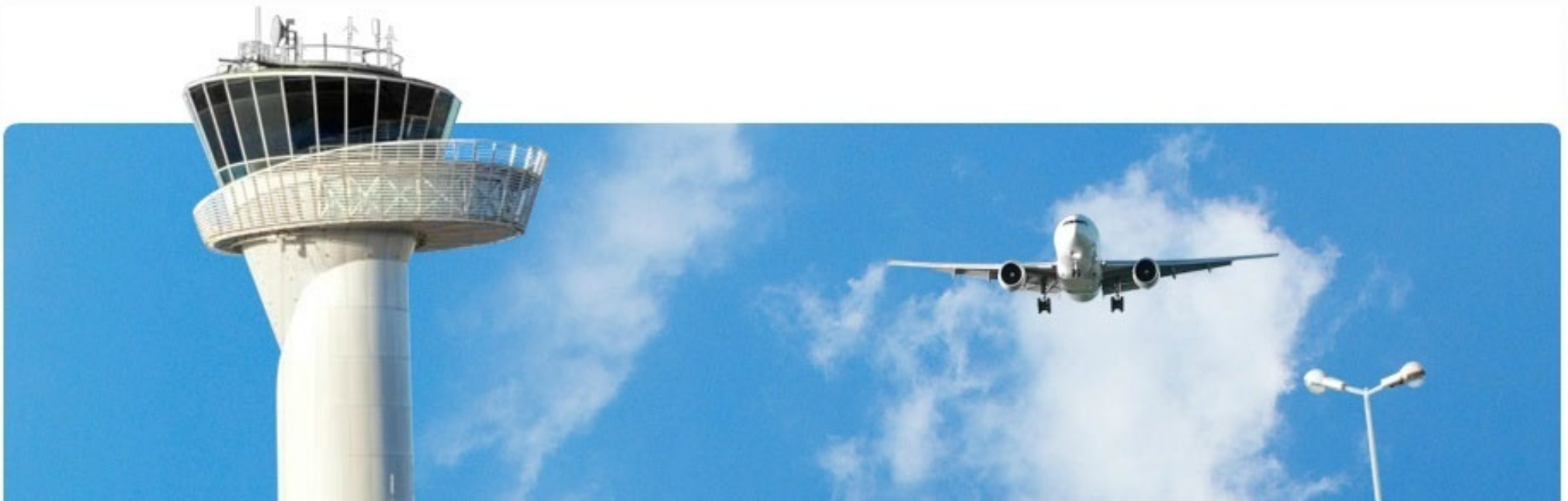

Models for Air Traffic Management

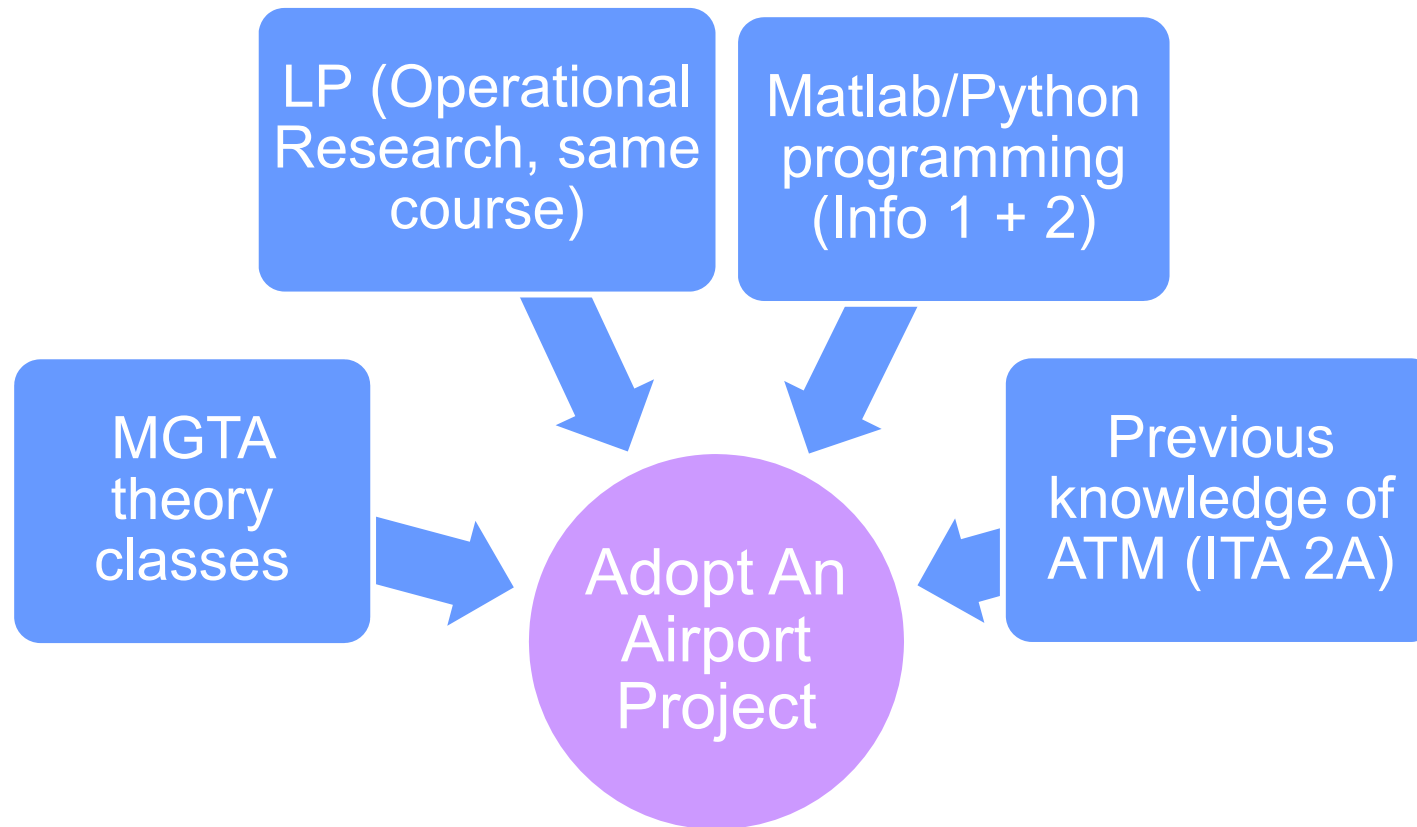


Adopt an Airport Project

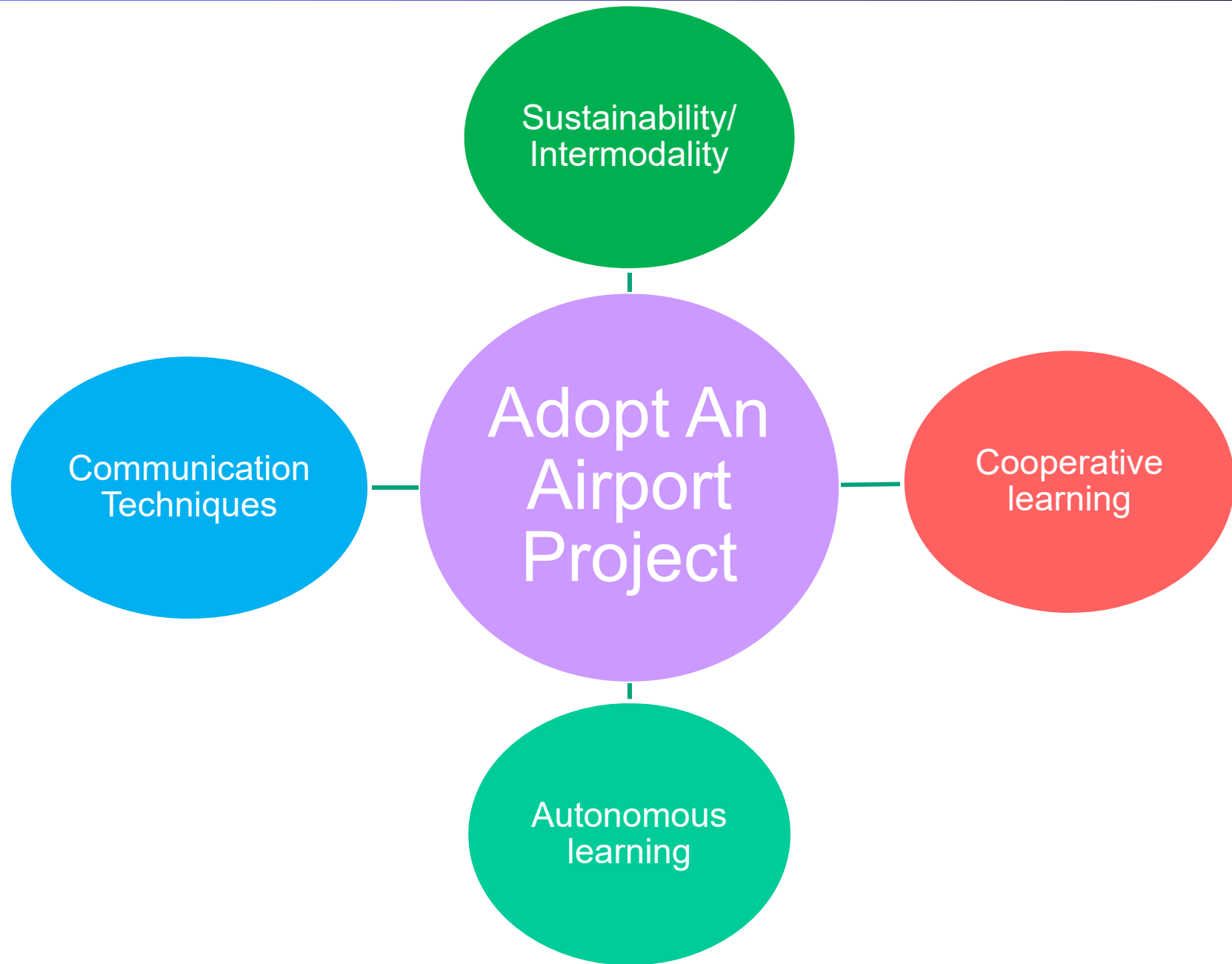
11/02/2026

Project objective

Given an chosen (adopted) European airport where traffic regulation is applied, **apply different ATFM methods** to it and study **relevant metrics** (number of flights affected, mean delay per flight, air vs ground delay, total delay, standard deviation of delay, extra cost, emissions of delay, etc.)



Project objective



Project index

WP1

- Generate data and define regulation on your airport

WP2

- Apply a GDP

WP3

- Apply a GHP

WP4

- Consider intermodality

WP5

- Analyze results, compare and evaluate methods, and conclude

Project planning

Weeks:	2	3	4	5	6	7	8	9	10	11
WP1 (Traffic+Reg)										
WP2 (GDP)										
1p progress report										
WP3 (GHP)										
WP4 (InterModal)										
WP5										
Oral test										
Deliver paper										

1 page progress report (16/03)

One page progress report giving main results of delays (KPIs) for the regulation defined, and the GDP solution

Deliverable: report due on May 4th (10 pages max.)

Two templates will be available (Word and Latex), make sure to respect basic rules of format (Figures and Tables captions, references, etc.)

Oral test (week 14: 26/05 and 29/05)

Oral test (in English) with short questions on the tasks you were involved in (contribution of group members will be delivered shortly before)

Continuous following of the work

Once a week, one member of the group will present to the professor the main results obtained by the group during the past week as well as their doubts. This group *leader/spoke(wo)man* should be the WP responsible (and thus change along the project)

From week 5, both MGTA classes will be dedicated to the project and longer consults and interviews will be possible.

References

1 class session (around week 8) will be dedicated to talk about references, databases, AI, correct use of citations to avoid plagiarism, etc.

Evaluation (project counts for 36% of MGTA total grade)

Weekly updates, participation to classes, stick to planning: 10%

One page progress report 5%

Individual oral test: 15%

Final report 70%

Accuracy and presentation of results, justification of choices made, quality of references used, originality of the results (graphs, chosen metrics, etc.) and **above all analysis of the results** presented will be (between others) some of the evaluation criteria

General advice

- **be flexible**: some data may change along the project, you may realize the regulation is not “constraining enough” and there is hardly any delay, or you may need to add an hour of study, or change the capacity, etc. Program in such a way (**use variables!**) that it will be no problem if you have to change a parameter and re-apply all methods
- **justify your choices**: the report should focus on the choice you had to make, or how you solved some challenges, not on the obvious parts
- **analysis**: invert enough time on **interpretation of results / comparison of methods/conclusions**, this is where good part of the grade will go
- **plan well**: try to stick to the chronogram, and you will have a smooth end of the project

General advice

- **Team work**: divide well the work: you need to properly **share tasks** **between group members** so that you progress fast enough, at the end of the project, you will need to deliver the **contributions of each member** (during the oral test, you will be evaluated on the tasks where you were more involved)
- Either **Matlab** (<https://serveistic.upc.edu/ca/distsoft/el-servei/programari-per-a-estudiantsi-professors#section-4>) or **Python** can be used.
- **AI/ChatGPT: its use must be cited in the report: see** <https://guies.biblioteca.upc.edu/com-citar/mitjans-digitalis#s-lib-ctab-16047452-5>

General advice

+ we can also answer any doubt you will have → ask us!

We will not solve code problems during office hours (nor by email), but we will help you with conceptual problems.

**WP.0: Form your group and send an email (to Adeline or David) with group members and airport name
→ Before 23rd of February**

WP.1: Generate data and define regulation on your airport

Project planning

Weeks:	2	3	4	5	6	7	8	9	10
WP1 (Traffic+Reg)									
WP2 (GDP)									
1p progress report									
WP3 (GHP)									
WP4 (InterModal)									
WP5									
Deliver paper									

WP.1: Generate traffic data at your airport

You have been assigned an airport (FRA, MUC, LHR, AMS, BCN, MAD, CDG, FCO (Rome), ZRH). Data is given at your chosen airport for Aug 10th, 2025 (all day operations):

1. For each flight at your airport you will thus directly have:
 - a. its flight number (ARCID),
 - b. the aircraft type (ATYP),
 - c. its departure airport, ICAO code (ADEP),
 - d. its arrival airport, ICAO code (ADES),
 - e. its registration mark (RM),
 - f. its reference flight level (ARF),
 - g. its estimated time of departure in UTC (ETD). Note that if it is larger than the ETA, it means the flights left the previous day
 - h. its taxi time “out” (TT),
 - i. its estimated time of arrival in UTC (ETA).

WP.1: Generate traffic data at your airport

2. For each flight arriving at your airport, you need to compute:
 - a. **aircraft category and available seats** (obtained from aircraft models): you can use the fleet_cat_seat.csv file, and for missing models use the new RECAT-EU categories (between A, the largest and F, the smallest), see for instance:
<https://www.easa.europa.eu/en/downloads/117238/en>
 - b. **flight distance**: it can be calculated from ETD & ETA to get the flight time, resting taxi time (out is given, in can be estimated):
<https://www.eurocontrol.int/publication/taxi-times-summer-2024>)
and then applying an average speed for the aircraft category,
 - c. whether the flight is **from outside ECAC** (<https://www.ecac-ceac.org/about-ecac/member-states>), you can filter by the ICAO origin airport code (LF is France: within ECAC, UU is Russia: outside ECAC, etc.)
 - d. **flight airline** (from the flight number)

3. Study your airport arrival capacity:

<https://www.eurocontrol.int/publication/technical-note-airport-capacity-imbalance>.

- a. Determine **nominal (AAR)** capacity from the EUROCONTROL document and set a **reduced (PAAR)** arrival capacity (consider for example very bad weather conditions that would greatly reduce the capacity)
- b. Calculate the corresponding size of the time slots **t** . These slots will be used for computing regulations delays, they do not necessary correspond to ATFM slots.

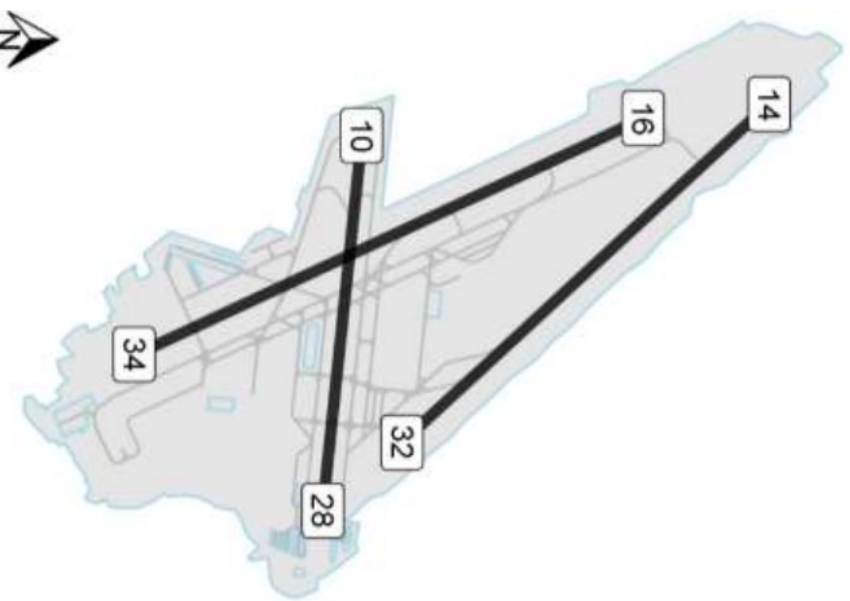
You can choose to have integer or not slots, and you can have 1 or 2 flights by slot depending on the runway configuration.

WP.1: Capacity study

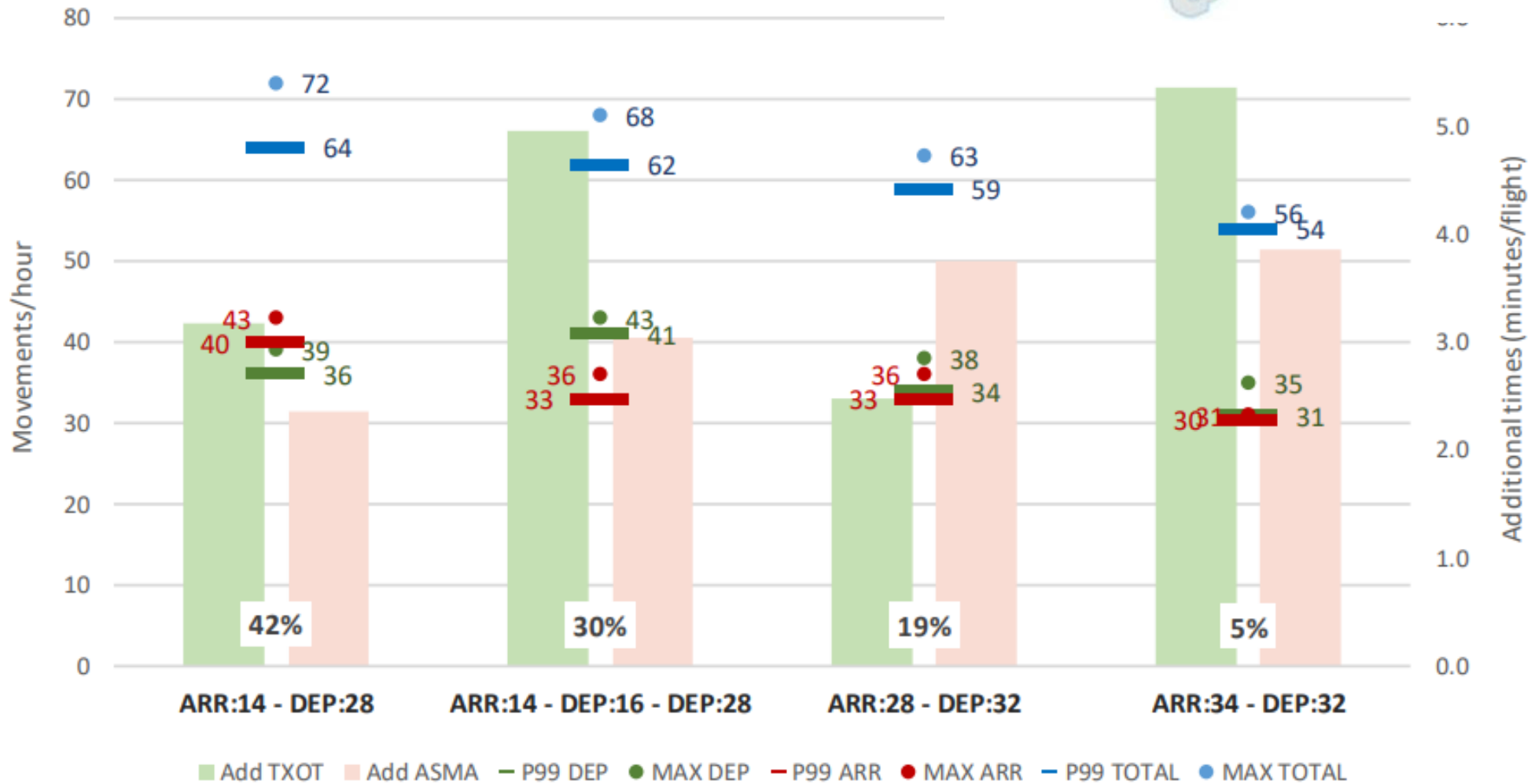


From the [AIRPORT CAPACITY
IMBALANCE STUDY, 2020]

ZRH example:



LSZH (ZRH) - Zürich



4. Define the hypothesis of your **regulation**, namely:
- **file time** (H_{file}): when the regulation is set and the regulated slots are sent to the airline
 - **starting time of reduced capacity** (H_{start}),
 - **end time of reduced capacity** (H_{end})

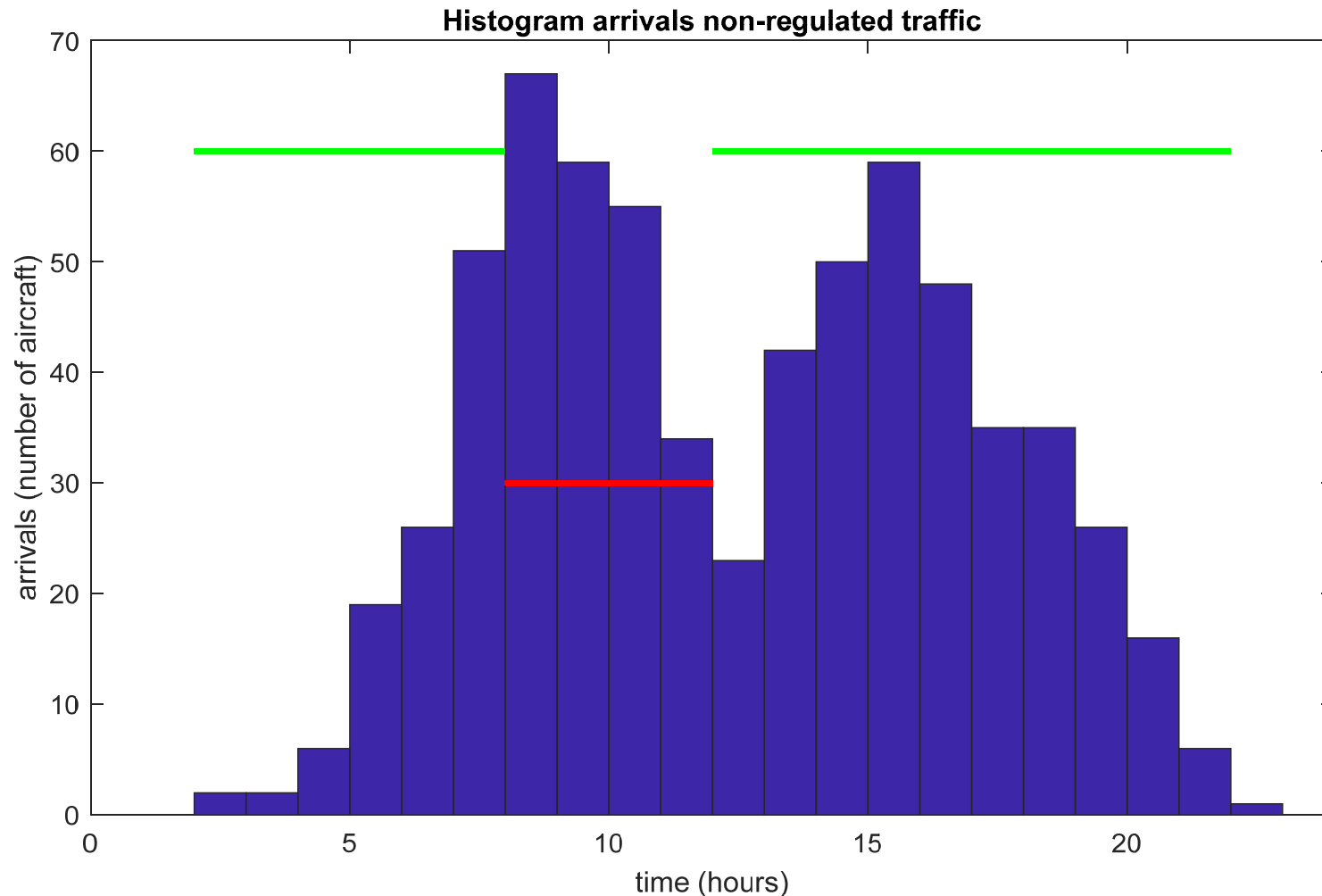
Please consider **at least 5 hours of reduced capacity** (between H_{start} and H_{end})

These parameters have to be very clearly indicated in your final report

5. After that, calculate **end time of regulation** (H_{noReg}), see next slides

WP1: Traffic demand data

Example of 24h simulated arrival demand at Logan International Airport (BOS)



Nominal capacity has been plotted as a green line and reduced capacity as a red line.

WP.1: Regulation definition

Program a function that computes the aggregate demand and capacities, the minimum total delay needed, and the hour when the regulation is no longer needed.

Suggested inputs: ETA, Hstart, Hend, PAAR, AAR

Suggested outputs: HNoReg, delay

This function will compute the aggregate demand as a function of time and generate a plot with this data.

WP.1: Regulation definition

Use a matrix "AggregateDemand" to compute the aggregate demand. The elements in `AggregateDemand(:,1)` will be the time in minutes and the elements in `AggregateDemand(:,2)` will be the number of aircraft (check [cumulative sum function](#) in Matlab: `cumsum`).

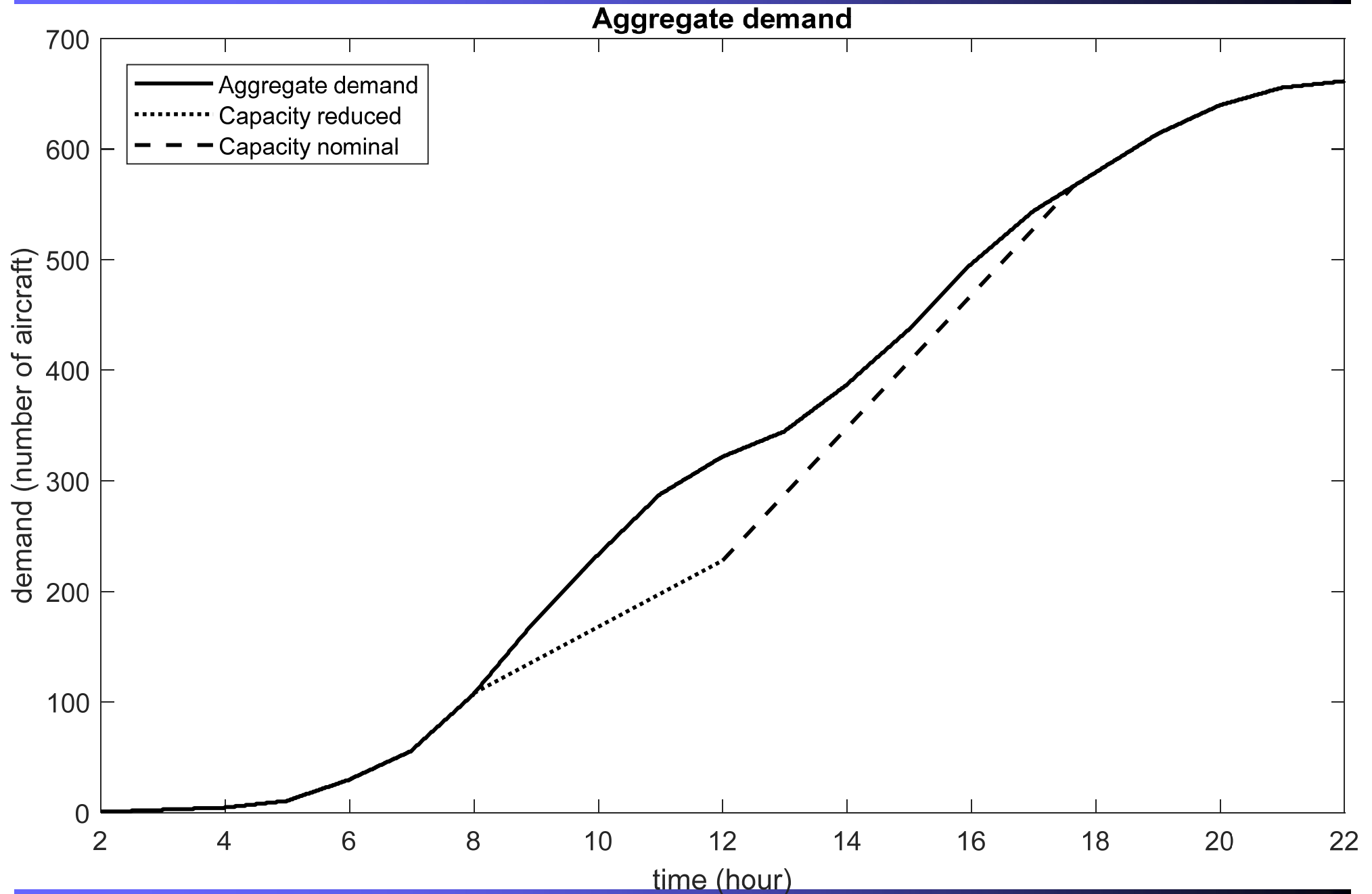
Example: ETA = [... ...; AggregateDemand= [... ...;

...	490	120;
491;	491	121;
493;	492	121;
493;	493	123;
...]	...	...]

means that at time 490 (8:10 am) the aggregate demand is of 120 arrivals. Do an entry in the matrix for each minute between the first and the last ETA of the arrival demand within the regulated period.

Use same idea to compute the aggregate capacity of regulated period.

WP.1: Regulation definition



WP.1: Regulation definition

Minimum total delay = difference between demand and capacity: distance between the continuous and discontinuous lines.

This delay only depends on the regulation (not on how you solve the regulation).

Check whether the average delay is large enough ($>45\text{min}$) so that the simulated regulation is interesting enough to study

Note that the results and the plot are only examples, with your data the plots will be different.

WP.1: Regulation definition

Program a function computing the slots of the regulation:

Suggested inputs: `Hstart`, `Hend`, `HNoReg`, `PAAR`, `AAR`

function computes the matrix of slots **ONLY** from `Hstart` to `HNoReg`

Each slot line information should include: **initial time of the slot**, **ID¹** of the assigned flight to that slot, and airline to which the slot has been assigned, both initially set to zero.

Size of the slots will vary according to the reduced (`PAAR`) and nominal capacity (`AAR`) of the airport.

¹ **flight ID** should be taken as its **original position in the ETA list** of flights

Example of slots:

```
slots = [ ...      ...  ...
          648      0   0
          650      0   0
          652      0   0
          654      0   0
          656      0   0
          658      0   0
          660      0   0
          661      0   0
          662      0   0
          663      0   0
          664      0   0
          665      0   0
          666      0   0
          ...      ...  ... ]
```

Note: Consider that when assigning the slots some of them might not be used (though not common during a regulation)

Therefore, you might consider adding some extra slots after HNoReg for some flights that might need them.

Once again you can use total minutes for slots also instead of hour minutes.

WP1: Performance assessment: Introduction

SESAR looking to establish **performance-driven network management culture** (evolution of current process, mainly focused on **delay reduction**)

Requires agreed **set of performance indicators** to be used for real time performance monitoring, trade-off analysis, and what-if impact assessments

Indicators need to be **dynamically** selected according to network geographic interest scales and specific time horizons

Identify **common local, regional, and sub-regional network performance indicators** and their relationship to a selected set of network congestion mitigation strategies

[SESAR Solution, 2019]

2015 Edition of the European ATM Master Plan: www.atmmasterplan.eu/exec/overview/performance-ambitions

Key performance area	SES high-level goals 2005	Key performance indicator	Performance ambition vs. baseline			
			Baseline value (2012)	Ambition value (2035)	Absolute improvement	Relative improvement
 Capacity	Enable 3-fold increase in ATM capacity	Departure delay⁴ , min/dep	9.5 min	6.5-8.5 min	1-3 min	10-30%
		IFR movements at most congested airports⁵ , million	4 million	4.2-4.4 million	0.2-0.4 million	5-10%
		Network throughput IFR flights⁵ , million	9.7 million	~15.7 million	~6.0 million	~60%
		Network throughput IFR flight hours⁵ , million	15.2 million	~26.7 million	~11.5 million	~75%
 Cost efficiency	Reduced ATM services unit costs by 50% or more	Gate-to-gate direct ANS cost per flight¹ , EUR(2012)	EUR 960	EUR 580-670	EUR 290-380	30-40%
		Gate-to-gate fuel burn per flight , kg/flight	5280 kg	4780-5030 kg	250-500 kg	5-10%
 Operational efficiency		Additional gate-to-gate flight time per flight² , min/flight	8.2 min	3.7-4.1 min	4.1-4.5 min	50-55%
		Within the: Gate-to-gate flight time per flight ³ , min/flight	(111 min)	(116 min)		
 Environment	Enable 10% reduction in the effects flights have on the environment	Gate-to-gate CO₂ emissions , tonnes/flight	16.6 tonnes	15-15.8 tonnes	0.8-1.6 tonnes	5-10%
		Accidents with direct ATM contribution⁶ , #/year Includes in-flight accidents as well as accidents during surface movement (during taxi and on the runway)	0.7 (long-term average)	no ATM related accidents	0.7	100%
 Security		ATM related security incidents resulting in traffic disruptions	unknown	no significant disruption due to cyber-security vulnerabilities	unknown	-

1 Unit rate savings will be larger because the average number of Service Units per flight continues to increase.

2 "Additional" means the average flight time extension caused by ATM inefficiencies.

3 Average flight time increases because the number of long-distance flights is forecast to grow faster than the number of short-distance flights.

4 All primary and secondary (reactionary) delay, including ATM and non-ATM causes.

5 Includes all non-segregated unmanned traffic flying IFR, but not the drone traffic flying in airspace below 500 feet or the new entrants flying above FL 600

6 In accordance with the PRR definition: where at least one ATM event or item was judged to be DIRECTLY in the causal chain of events leading to the accident. Without that ATM event, it is considered that the accident would not have happened.

Introduction

2025 European ATM Master Plan: https://www.sesarju.eu/MasterPlan2025_SupportingDocuments

Performance impact (KPA)	Unit	Reference year (2023) ^(*)	CP1 (up to 2030)	CP1 + Phase C	Phase D	Expected impact by 2050
Airspace capacity (en-route and TMA)	%	8.5 million flights ^(*)	+ 34 %	+ 60 %	+ 40 % + 80 %	+ 100 % + 140 %
Airspace capacity	%	17.9 million movements ^(**)	–	+ 15 %	+ 1 % + 5 %	+ 16% + 20 %
Environment (fuel reduction)	kg / flight	6 400	– 22	– 109	– 491	– 600
	%		– 0.3 %	– 1.6 %	– 7.7 %	– 9.3 %
Passenger time saving (departure punctuality)	minute / flight	18	–	– 0.9	– 6.1 – 8.1	– 7 – 9
Cost-efficiency (air navigation services cost reduction)	EUR / flight	1 077	– 26	– 164	– 54	– 209

WP1: Performance assessment: Terminology

Key Performance Area (KPA)

A way of categorising performance subjects related to high level ambitions and expectations

Key Performance Indicator (KPI)

Current/past and expected future performance, & actual progress in achieving performance objectives **quantitatively expressed by means of indicators**

Indicators need to correctly express intention of associated performance objective: since they support objectives, they should not be defined without having a specific performance objective in mind

Indicators are not often directly measured. They are calculated from **supporting metrics** according to clearly defined formulas

[PJ19.04]

Selection of KPAs uses as reference set of KPAs defined in ICAO framework with refinements to support SESAR requirements

1. Safety
2. Security (e.g.: [EUROCONTROL Data Snapshot Ransomware groups target aviation's supply chain](#))
3. Environment
4. Capacity
5. Operational Efficiency
6. Predictability
7. Cost effectiveness
8. Flexibility
9. Access and Equity

[PJ19.04]

WP1: identify KPAs/KPIs (during all project)

The results of this project will have to be presented as KPIs (that belong to a specified KPA).

During each WP, you need to **identify what KPIs can be computed** (and classify them under KPAs).

As seen with the example of the update of the ATM master plan, KPA definition can be flexible.

For instance, in WP1, you are asked to compute the total delay of the regulation [min] or [h], that would be a KPI, and that can be considered under the “Capacity” KPA.

WP.2: Apply Ground Delay Program

Project planning

Weeks:	2	3	4	5	6	7	8	9	10
WP1 (Traffic+Reg)									
WP2 (GDP)									
1p progress report									
WP3 (GHP)									
WP4 (InterModal)									
WP5									
Deliver paper									

WP.2: Apply Ground Delay Program

Apply a [Ground Delay program \(see theory class\)](#) for this regulation affecting arrival flights between H_{start} and H_{noReg} (controlled flights)

Compute the corresponding delay for each flight and other interesting KPIs (total air/ground delay, mean air/ground delay per flight, relative air/ground delay standard deviation, max air/ground delay, CO₂ emissions due to delay, on-time performance, etc.).

Program a function computing the list of not affected, excluded and controlled flights

Suggested inputs: `ETA`, `ETD`, `Distances`, `International`, `Hfile`, `Hstart`, `HNoReg`, `radius`

Suggested outputs: `NotAffected`, `ExemptRadius`, `ExemptInternational`, `ExemptFlying`, `Exempt`, `Controlled`

From the arrival demand at the airport, this function computes which flights:

- are **not affected** by the GDP because their ETA is earlier than the initial time of the GDP (`Hstart`) or because their ETA is after the hour when the regulation is not active (`HNoReg`),

Program a function computing the list of not affected, excluded and controlled flights

From the arrival demand at the airport, this function computes which flights:

- are **exempt** because their origin airport is further than the **radius** of the GDP
- are **exempt** because are already **flying** when the GDP is defined. That is, their ETD is earlier than the file time of the regulation (Hfile).

Consider a margin (**30 min for example**) to exempt flights ready to depart at Hfile (already boarded, door closed, etc.)

- are **exempt** because they come from **international** airports
- are controlled (included in the GDP)

Each set of flights is a vector containing the ID of each flight.

WP.2: GDP: Assign slots and compute delays

Program a function assigning each flight to a slot and computing and the delay assigned and its kind (air or ground)

Suggested inputs: slots, Controlled, Exempt, ETA, ETD

Suggested outputs: slots, GroundDelay, AirDelay

This function will assign slots to the flights that are arriving to the airport, assigning slots first to exempt flights, then to controlled ones (always respecting that a flight cannot arrive before its original ETA).

AirDelay is considered as the delay to apply to all **exempt** flights

GroundDelay is considered as the delay to apply to all **controlled** flights

WP.2: GDP: Assign slots and compute delays

Program a function assigning each flight to a slot and computing and the delay assigned and its kind (air or ground)

```
GroundDelay = [ ...  ...  
                97    9  
                101   14  
                104   30  
                106   32  
                ...   ...]
```

Both delay matrices relate the ID of the flight with the amount of delay that has been assigned to that flight. From the previous example, flight number 97 has been assigned 9 minutes of ground delay.

WP.2: GDP: Assign slots and compute delays

Program a function assigning each flight to a slot and computing and the delay assigned and its kind (air or ground)

The matrix of slots will be completed by adding the ID of the flight assigned to each of the slots. See that, depending on the demand, some slots might remain unused (not very likely when demand is high, but not impossible), like in the following example (slot at 640, i.e. from 10h40 to 10h42):

```
slots = [...  ...  ...  
          630      95  A1  
          632      96  A2  
          634      97  A1  
          636      98  A3  
          638      99  A1  
          640       0   0  
          ...      ... ...]
```

Program a function computing the CTA of the flights

Compute the CTA of all flights from the original ETA list of traffic and the delay assigned. CTA is the same as ETA but with the delay added to the flights that have to do delay.

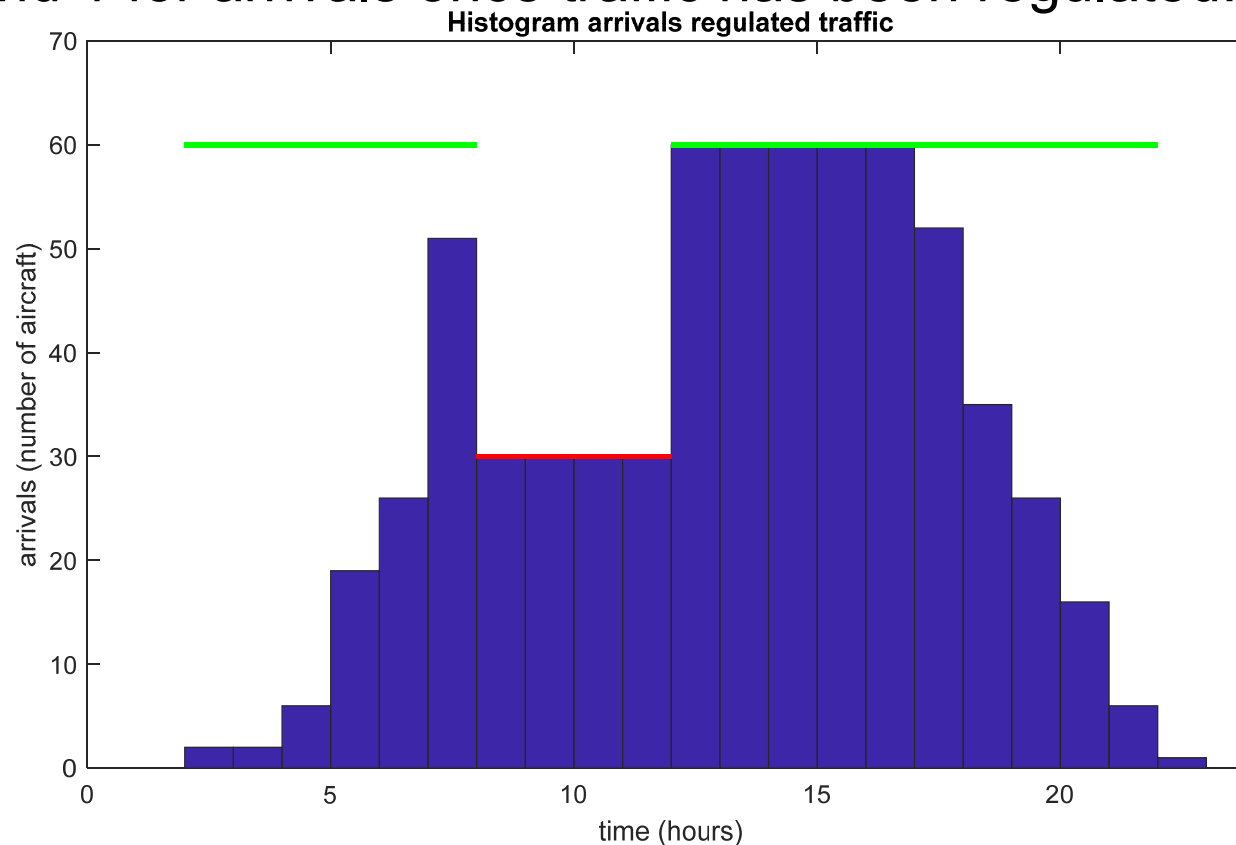
Alternatively, CTA can be obtained in your function `assignSlotsGDP`

At this stage, ATCSCC would send the controlled slots of arrival (and departure) to the airline.

WP.2: GDP: controlled histogram

Results: plot histograms

Generate two histograms, one for arrival traffic before regulation (see slide 20) and 1 for arrivals once traffic has been regulated:



This serves to help checking if your GDP is correctly implemented

Compute the GDP unrecoverable delay

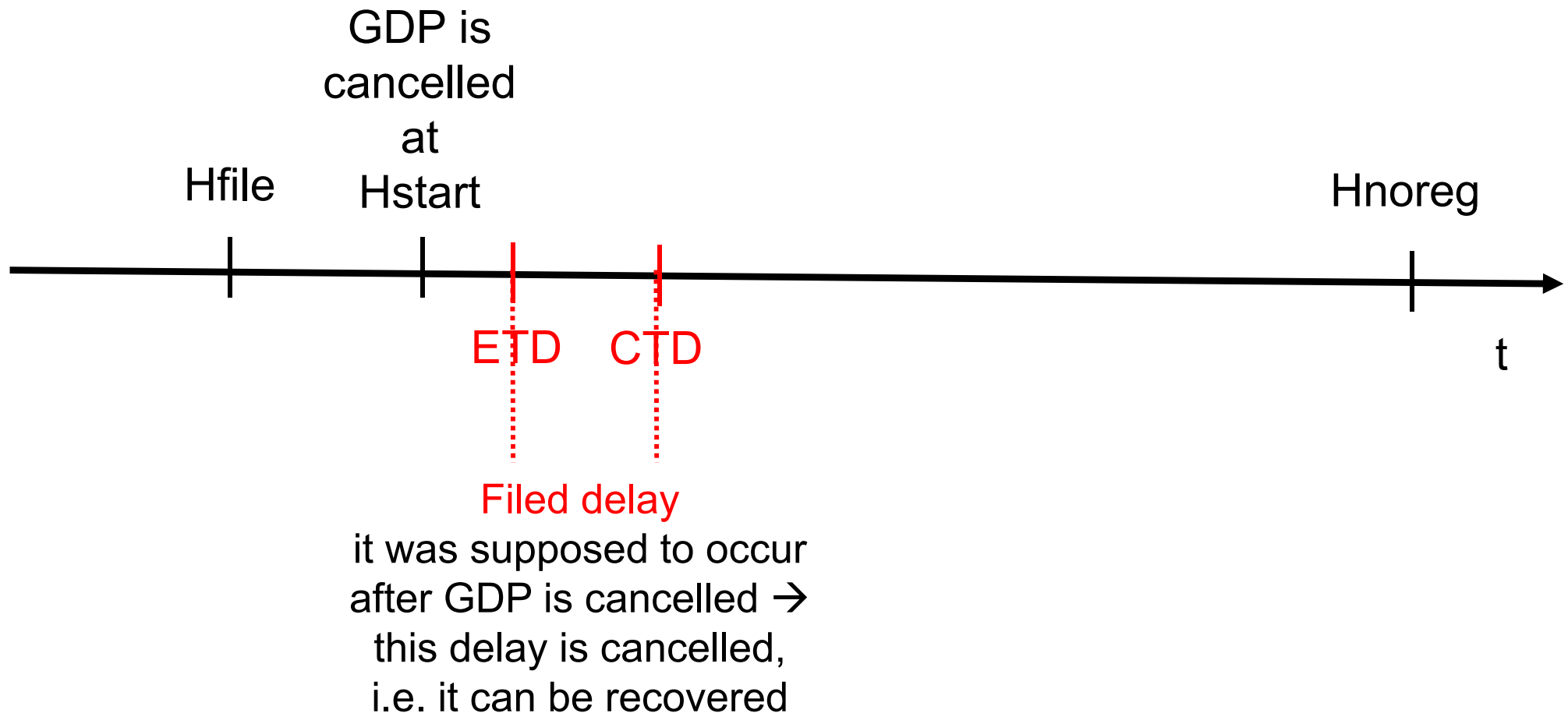
Make the assumption that the GDP is cancelled at H_{start} , and thus no reduced capacity is needed anymore (nominal capacity).

You do not need to assign slots again, but from the GroundDelay previously computed, calculate the unrecoverable delay: delay that has already been done, even though the GDP is cancelled at H_{start} .

WP.2: GDP: unrecoverable delay

Compute the GDP unrecoverable delay

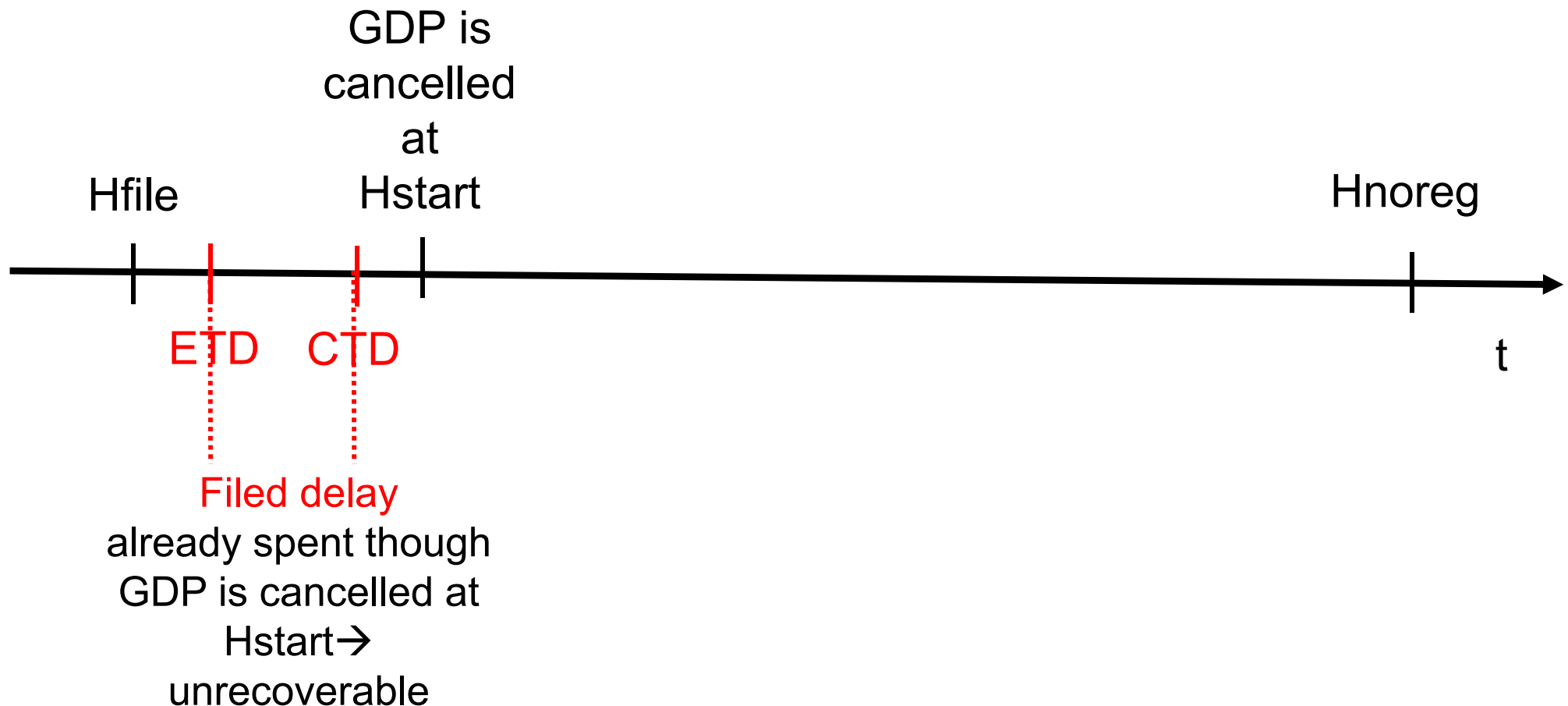
Case 1: $ETD > Hstart$ → no unrecoverable delay, delay completely cancelled



WP.2: GDP: unrecoverable delay

Compute the GDP unrecoverable delay

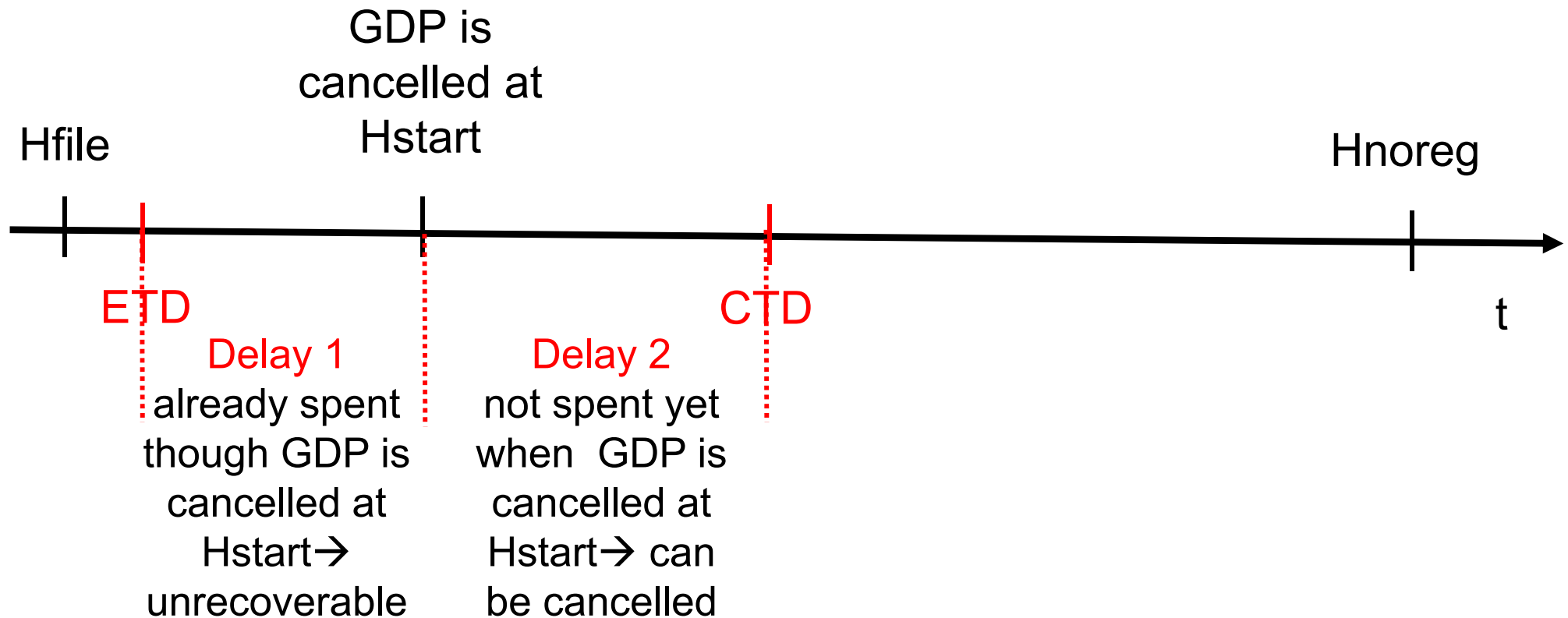
Case 2: $CTD < Hstart \rightarrow$ all assigned delay is unrecoverable delay, since it was already done when GDP is cancelled, unrecoverable delay = $CTD - ETD$



WP.2: GDP: unrecoverable delay

Compute the GDP unrecoverable delay

Case 3: $ETD < Hstart$ & $CTD > Hstart \rightarrow$ 1st part of assigned delay is unrecoverable, since it was already done when GDP is cancelled, 2nd part can be recovered, supposing flight gets to leave at $Hstart$ (ideal hypothesis)



WP.2: CO₂ emissions due to delay

Emissions due to **air delay**:

Use the emissions_fuel_estimation model described in the following paper: [Analytical Models of Flight Fuel Consumption and Non-CO2 Emissions as a Function of Aircraft Capacity](#), and available on Github: https://github.com/luis-uow/emissions_fuel_estimation

Given input of: flight distance and available seats, it will return: fuel consumption, CO₂ emissions (recalling from Q and SEA that 1kg of aviation fuel burnt generates 3.16 kg of CO₂ [IATA Carbon Offset Program](#)) and other non-CO₂ emissions per seat and kilometer (ASK).

From the emissions/ASK, and using the average velocity previously defined, you can deduce the emissions per minute of each flight and then the **emissions due to air delay during the regulation**.

Some flights may be outside of the range of the model, you will have to decide how to compute their emissions.

WP.2: CO₂ emissions due to delay

Emissions due to **ground delay**:

Look for references for fuel consumption “at gate”.

If possible, use one value of at gate fuel consumption for each aircraft category (A to F).

Once you have the ground delay fuel consumption, calculate **the CO₂ emissions due to ground delay during the regulation** (again, 1kg of aviation fuel burnt generates 3.16 kg of CO₂)

Finally compute the total CO₂ emissions due to delay (air + ground)

WP.2: GDP effect of parameters

Study the effects of changing the GDP parameters

Study the effect of the *radius of exemption* and *file time* on several KPIs.

Generate plots showing the trade-off between air delay, CO₂ emissions due to ground and air delay, and unrecoverable delay (and any other KPI that you think can be of interest) as functions of the file time and the radius of exemption.

Use heatmap plots to show the effect of both HFile and R on the same graph for each KPI.

From these plots define an optimal value for R and HFile.

Using R and Hfile previously chosen, simulate now that one airline (one with lots of flights at your airport) decides to cancel all flights with high delays (for instance > 2 or $3h$ depending on your airport)

Perform the second slot-assignation of GDP (after cancellations)

1. Identified flights to cancel and eliminate them from the controlled slots
2. Apply substitution for the chosen airline (maintaining original order of flights)
3. Once this airline has used all its emptied slots, apply compression in order to fill all slots.

Give the result of the new assignment of the aircraft to the slots and the delay assigned to each flight, for instance in a graphical way

Identify interesting KPIs for this part (number of cancelled flights, decrease of delay in remaining flights for this airline, etc.)

Importance of performance measure of the GDP in your report:

Identify KPAs and compute and analyze at least the following KPIs:

- total, maximum and average delay that are realized as ground delay and as air delay
- relative standard deviation of ground delay and of air delay
- CO₂ emissions due to ground delay and to air delay
- other KPIs such as: number of flights delayed, OTP: number of flights delayed with less than 15 min, etc.

Again, any other plots/results that you find interesting should be added to your report and will be valorized

WP.2: A remark about total delay

We recall that the total delay is constant for any method applied and depends only on the hypothesis of the regulation

Nevertheless some groups may experience very small variations of total delay (a few minutes). This is due to the change of order of the flights when your slots are of several minutes as illustrated in the following example, where F2 is an exempt flight when applying GDP:

Flight	ETA	Slots	RBS		GDP	
			Assignment	Delay	Assignment	Delay
F1	8.00	8.00	F1	0	F2	0
F2*	8.01	8.02	F2	1	F1	2
F3	8.02	8.04	F3	2	F3	2
F4	8.03	8.06	F4	3	F4	3
Total delay:				6		7

Since the variation of delay is small (usually $\ll 1\%$) with respect to the total of delay, you can neglect this difference in your conclusions

MGTA references

- [AIRPORT CAPACITY IMBALANCE STUDY, 2020] PERFORMANCE REVIEW COMMISSION, EUROCONTROL <https://www.eurocontrol.int/publication/technical-note-airport-capacity-imbalance>, accessed on 10/07/2023
- [PJ19.04] PJ19.04: Performance Framework (2019) https://www.sesarju.eu/sites/default/files/documents/awards2021/SESAR%20Performance%20Framework%20ed_%2001_00_01%20-%202019.pdf, accessed 10/07/2023
- [SESAR Solution, 2019] SESAR SOLUTIONS CATALOGUE 2019 Third edition www.sesarju.eu/sites/default/files/documents/reports/SESAR_Solutions_Catalogue_2019_web.pdf, accessed 10/07/2023